

**Name:** Jazlynn, Bryce, Kam, Quest      **Date:** 2/9/26      **Class:** 8

# MyDesign Portfolio

## *Element D Initial Template*

### How to use this template

Teachers: You may customize this document before providing it to students.

Students: The template below is one way to present your design ideas, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your own design ideas, be sure to delete the instructions.

### Guidance before filling out the template on the next page:

Because you will iterate on your design multiple times, you will need to create multiple copies of these sections. Think of Element D as a diary that you will date and keep adding to throughout your work until your final iteration. It's critical that iteration shines through in this Element.

You should begin with many design ideas that reflect the individual ideas of your group members and then the group ideas. Number these ideas and include a date, description, drawing, and details for each.

After you have created an initial set of ideas, document your decision making on which design to include, ideally with a design matrix. Create a plan of work, ideally using a Gantt chart that may be housed in this document or in a different day-to-day document. Generate a design blueprint with enough details that someone else not familiar with your project could also build your design.

You will then proceed forward to build and test this design, returning to Element D afterwards when you are ready to create more design ideas. Fill out each of these steps again.

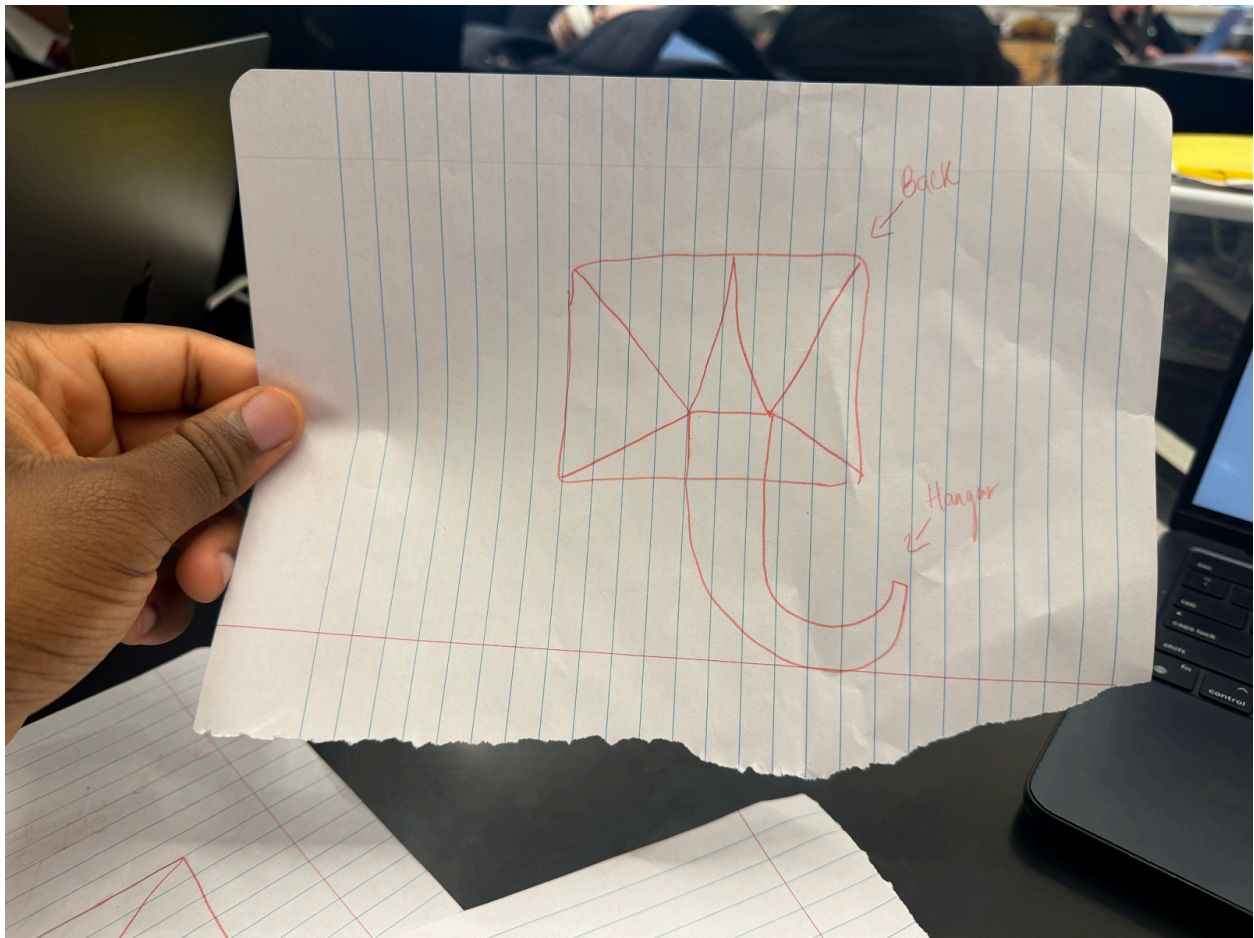
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## Element D: Design Concept Generation, Analysis, and Selection

### Design Idea Generation (D1)

This section focuses on the process of generating solutions. The initial rounds should include every idea you come up with but the subsequent rounds of ideation should lead to viable options. Each idea should include not only words, but also a picture. Use as many copies of this sub-template as needed.

#### Design Idea #1



Date: 2/11

Description: Use a semi flexible material to see if it helps

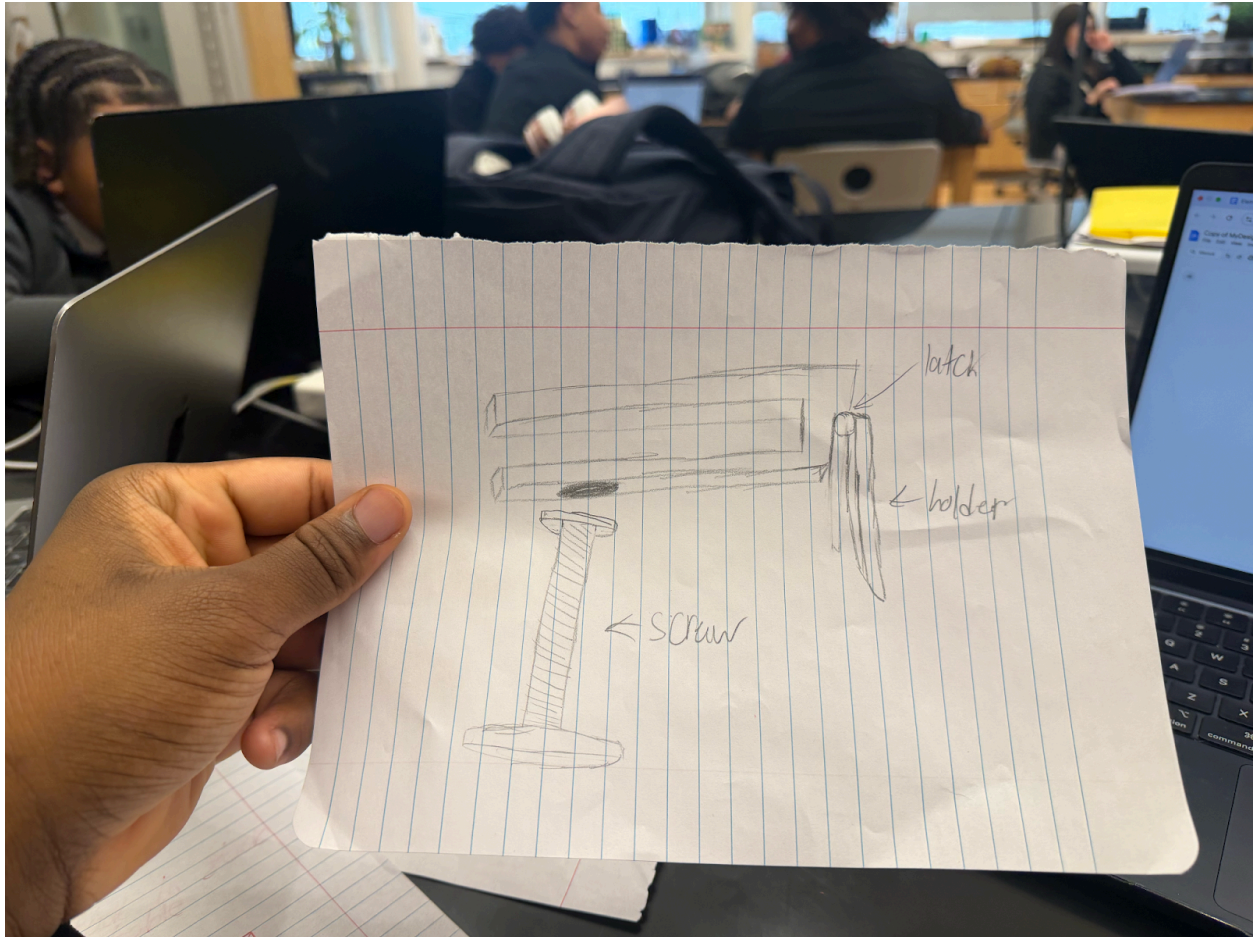
Drawing that includes distinguishing information such as scale, dimensions, and materials:

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*Design Idea #2*



Date: 2/11

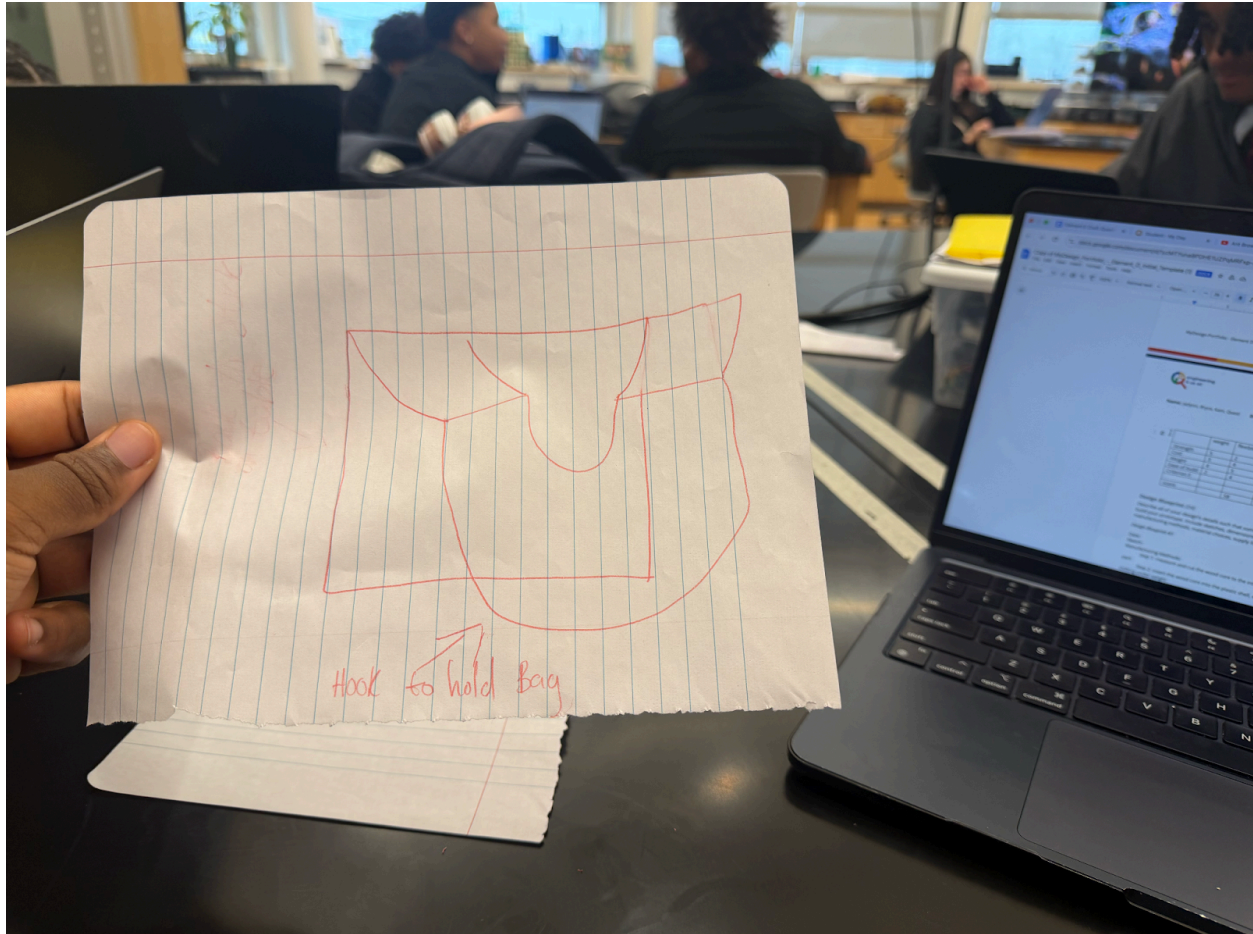
Description: Putting wood inside of the plastic so it can hold a lot and weigh little.

Drawing that includes distinguishing information such as scale, dimensions, and materials:



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### *Design Idea #3*



Date: 2/11

Description: Grow the 3d in size to see if it will help hold bigger weights.

Drawing that includes distinguishing information such as scale, dimensions, and materials:

### **Design Idea Comparison and Selection (D1, D2)**

When you've finished generating ideas, you must compare your design options according to all your design requirements. Incorporate the priority of those requirements when comparing design options. A decision matrix (perhaps a Pugh Matrix) is encouraged since it allows for systematic, weighted comparisons. If using weights in a decision matrix, be sure to explain and justify the weighting based on your design requirements. You may use the weighted design matrix template below, inserting your criteria and design options.



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|               | weight | flexible | Wood core | 3d scaled | Option 4 |
|---------------|--------|----------|-----------|-----------|----------|
| Strength      | 5      | 2        | 5         | 4         |          |
| Cost          | 3      | 4        | 2         | 3         |          |
| Weight        | 4      | 5        | 2         | 3         |          |
| Ease of build | 2      | 4        | 2         | 5         |          |
| Criterion E   |        |          |           |           |          |
|               |        |          |           |           |          |
| score         | 14     | 58       | 52        | 61        |          |

### Design Blueprint (D4)

Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

#### Design Blueprint #1

Date:

Sketch:

Manufacturing Methods:

Step 1: measure and cut the wood core to the internal dimensions of your plastic part

Step 2: insert the wood core into the plastic shell, ensuring a tight fit to prevent shifting under weight

Step 3: use something to create pilot holes through the plastic and into the wood , then drive the screws in to secure the assembly

Supply List and Costs:

| Item   | Cost each | # needed | Total |
|--------|-----------|----------|-------|
| wood   | \$10      | 1        | \$10  |
| screws | \$2.5     | 4        | \$10  |

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|                   |       |   |        |
|-------------------|-------|---|--------|
| plastic           | \$7.5 | 3 | \$22.5 |
| <b>Total Cost</b> | \$20  | 8 | \$42.5 |

### Plan of Action (D3)

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.